PROJECT PROPOSAL & TECHNICAL DESIGN SPECIFICATION

Industrial Custom Variable CC/CV DC-to-DC Charging Station System

Industry Partner: Sawant Automation, Alandi, Pune

CEO & Industry Lead: Mr. Dnyaneshwar Baliram Sawant

Academic Institution: Vishwakarma Institute of Technology (VIT), Pune

Faculty Project Guide: Dr. Vijay Gaikwad Sir

Student Team Lead: Chaitanya Giri

Project Target Timeline: 2 Months (8-Week Build Cycle)

1. EXECUTIVE SUMMARY & PROJECT OBJECTIVE

Electric Vehicle (EV) servicing and bench testing require versatile power sources capable of delivering precise, adjustable voltages and constant-current limitations. This project presents the complete engineering design and fabrication of an **Industrial Benchtop CC/CV DC-to-DC Charging Station** built around the AmiciSmart 1200W 20A DC-DC Step-Up Boost Converter.

The final product transitions an open PCB module into a rugged, workshop-grade diagnostic charging unit featuring a custom 3D-printed enclosure, external panel-mounted potentiometer controls, active auxiliary thermal management, reverse-polarity output protection, and real-time telemetry metering (V/A).

2. ENCLOSURE DESIGN & FINAL PRODUCT VISUALIZATION

Figure 1: 3D Visualization of the Custom Benchtop DC-DC Charging Station Enclosure with Front Controls and Side Cooling Panel.

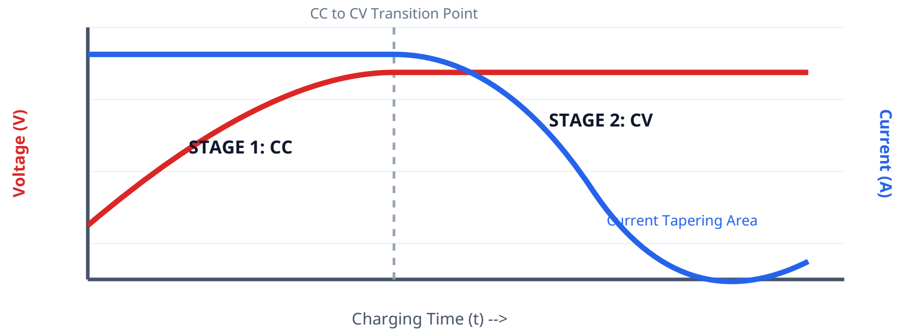
3. BATTERY CHARGING LOGIC & VOLTAGE/CURRENT DYNAMICS

The charging station operates on a standard two-stage **Constant Current / Constant Voltage (CC/CV)** charging protocol essential for Lithium-ion and Lead-Acid EV battery packs:

CC/CV Operational Stages:

- **Stage 1 - Constant Current (CC):** When a depleted battery is connected, internal resistance is low. The converter clamps the charging current to the user-defined panel limit I_{set} (e.g., 5A–10A) to prevent over-current stress. Voltage steadily rises.
- **Stage 2 - Constant Voltage (CV):** Once the battery reaches its target pack voltage V_{set} (e.g., 54.6V for a 13S Li-ion pack), the converter holds the voltage steady. As the battery internal potential rises, current naturally tapers down toward zero.

Figure 2: CC/CV Battery Charging Profile (Voltage vs. Tapering Current over Time)



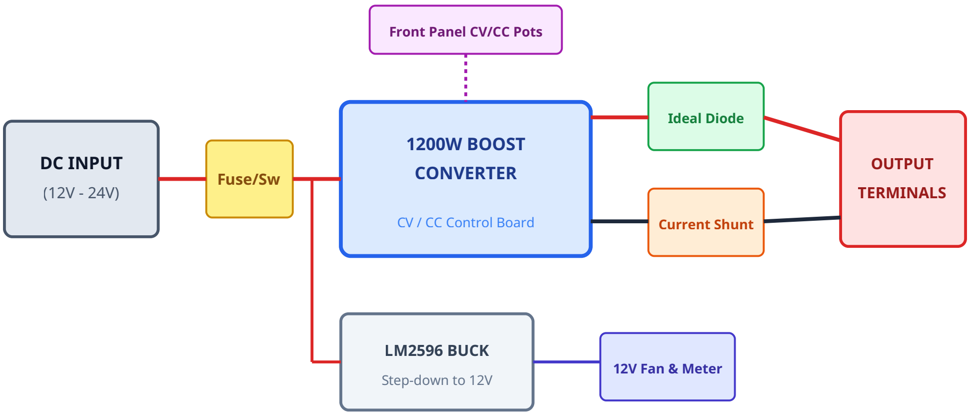
4. SYSTEM BILL OF MATERIALS (BOM) & TECHNICAL JUSTIFICATION

Component	Qty	Functional Role	Technical Specification
AmiciSmart 1200W Boost Converter	1	Core power step-up engine	Input 8–60V DC, Output 12–80V DC, Max 20A Input
Panel Rotary Potentiometers	2	External CV and CC tuning knobs	10kΩ / 100kΩ Linear Potentiometers (Panel Mount)
Dual Voltmeter / Ammeter	1	Real-time telemetry readout	0–100V, 50A/100A Dual LED Display
External Brass Current Shunt	1	Precision low-side current sensing	50A / 75mV (or 100A / 75mV) precision shunt bar
12V DC Auxiliary Cooling Fan	1	Forced thermal dissipation	80mm / 90mm High-CFM DC Fan with finger guard
LM2596 Step-Down Buck Module	1	Auxiliary 12V supply for fan & display	Input 4–40V, Output adjusted to stable 12.0V DC
Ideal High-Current Diode Module	1	Reverse-polarity battery back-feed protection	50A / 100V rated Anti-Backfeed Diode
Connectors & Interconnects	1 Set	High-current input/output interface	XT90 / Heavy-Duty Binding Posts, 10/12 AWG silicone wire

5. MASTER ELECTRICAL SCHEMATIC & INTERCONNECTION ARCHITECTURE

The complete system interconnection topology is mapped below, highlighting the auxiliary power isolation and low-side current shunt measurement architecture:

Figure 3: High-Current Power & Low-Voltage Signal Interconnection Block Diagram



6. FRONT & REAR PANEL LAYOUT MAPPING

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================================================================================
                            FRONT PANEL INTERFACE LAYOUT
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+------------------------------------------------------------------------------+
|  SAWANT AUTOMATION                                                           |
|  Alandi, Pune                                +------------------------------+ |
|                                              |    58.8 V  (Voltage Display) | |
|  PROJECT TEAM: Chaitanya Giri & Team         |                              | |
|  GUIDE: Dr. Vijay Gaikwad Sir                |    10.5 A  (Current Display) | |
|  INSTITUTE: VIT Pune                         +------------------------------+ |
|                                                    DUAL DIGITAL METER        |
|                                                                              |
|       ( O )                     ( O )                       [  I  ]          |
|     VOLTAGE                    CURRENT                       POWER           |
|     ADJUST                     ADJUST                        SWITCH          |
|  (Potentiometer)             (Potentiometer)                (Rocker)         |
|                                                                              |
|               ( RED )                 ( BLACK )                              |
|               (+) OUT                 (-) OUT                                |
|            BINDING POST              BINDING POST                            |
+------------------------------------------------------------------------------+
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7. PRECISION METER CALIBRATION PROCEDURE

Voltage Calibration:

1. Connect a calibrated Digital Multimeter (DMM) across the Output Binding Posts.
2. Turn on power and set Voltage Knob to mid-range (~54.6V).
3. Locate the **V-ADJ** potentiometer on the back of the panel meter.
4. Adjust until the panel display matches the external DMM voltage exactly.

Current Calibration:

1. Connect an adjustable high-power dummy load across the output terminals.
2. Set Current Knob to draw a steady 5.0A load.
3. Locate the **I-ADJ** potentiometer on the back of the display.
4. Adjust until the panel current matches the reference current reading.

8. OFFICIAL SPONSORSHIP REQUEST LETTER

Date: August 1, 2026

To,
Mr. Dnyaneshwar Baliram Sawant
Founder & CEO, Sawant Automation,
Gat No. 1205, Charholi Phata, Alandi - Markal Road, Alandi, Pune, Maharashtra – 412105

Subject: Request for Formal Project Sponsorship & Component Support for Industrial EV Charger Project

Respected Sir,

We extend our sincere gratitude for taking the time to interact with our team and for considering our project proposal under the industrial guidance of **Sawant Automation**.

Following our technical discussion, our team has formally initiated the development of a **Custom Variable CC/CV DC-to-DC Charging Station System** utilizing the 1200W Boost Converter module platform. The objective of this project is to build a complete, workshop-ready EV benchtop charging unit featuring a custom rugged enclosure, external CC/CV panel controls, active thermal cooling management, real-time telemetry metering, and safety protection circuits.

To ensure the successful execution, fabrication, and testing of this system within our 2-month target timeline, we kindly request Sawant Automation to formally sponsor this project and support us with the necessary raw components, module hardware, and testing facilities as required during the build phase.

Under the academic mentorship of our faculty guide, **Dr. Vijay Gaikwad Sir**, our team will ensure that the system is designed to industrial standards. Upon completion, all technical documentation, CAD models, wiring schematics, and test validation reports will be submitted directly to Sawant Automation.

We kindly request you to issue an official project acceptance letter confirming sponsorship and granting permission for our team to proceed.

Yours sincerely,	Under the Guidance of:
Student Project Team: 1. Chaitanya Giri (Team Lead) 2. [Team Member 2 Name] 3. [Team Member 3 Name] **Department:** Engineering **Institute:** VIT Pune	______________________________ **Dr. Vijay Gaikwad Sir** Faculty Project Guide, Department of Engineering